



Fundamentals of Programming: File Handling

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What is a file?



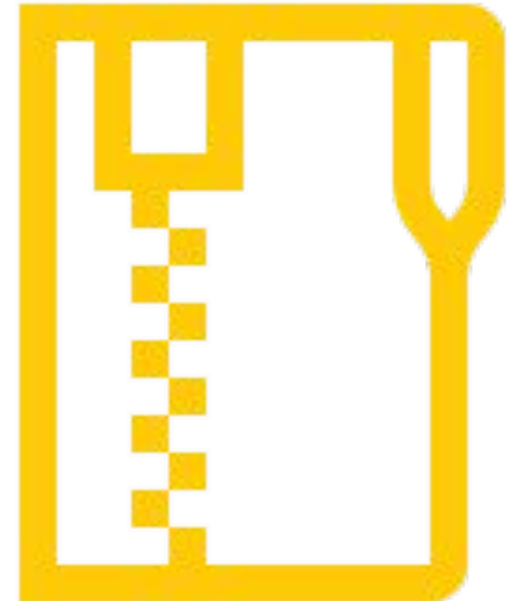
Unit of stored data that is identifiable by a unique name and stored in a filesystem.



It can contain various types of information, such as text, images, videos, executable code, or any other form of data.



Files can be of various types, but we will work with text files.



File Handling

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A mechanism to store the output of a program in a file, get the data in a program from a file.

Why Filing, when we have arrays/variables to store data:

Files help store these data permanently on a storage device.

Why we need it?

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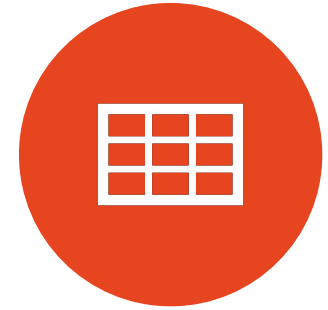
SAVING STUDENT
RECORDS, LOGS, AND
REPORTS



STORING PROGRAM
OUTPUT PERMANENTLY



READING
CONFIGURATION FILES



SIMPLE DATABASES IN
SMALL PROJECTS

Stream:

- ▶ Stream means flow of data, data can flow from console to the execution program, and from execution program to the console.

Fstream

- ▶ Header file, which gives access to file handling methods in C++
- ▶ Three classes of fstream:
 - ▶ ofstream
 - ▶ ifstream
 - ▶ fstream

fstream library

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ofstream:

Helps to write data to the file,
also known as output stream



ifstream:

Helps to read data from the
file, also called input stream



fstream:

Combination of ifstream and
ofstream, provides capability
of reading and writing a file

File operations in C++

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open() – This is used to open/create a file.

2

read() – This is used to read the data from the file.

3

write() – This is used to write new data to file.

4

close() – This is used to close the file.

Opening a file

- ▶ To read or enter data to a file, we need to open it first.

Syntax:

```
1 | open( FileName , Mode );
```

Here:

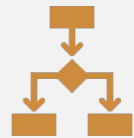
FileName – It denotes the name of file which has to be opened.

Mode – There different mode to open a file and it explained in this article.

Modes for a file



ios::in – Opens a file for **reading**.



ios::out – Opens a file for **writing** (overwrites if it exists).



ios::app – Opens a file for **appending**, writing at the **end** without erasing existing content.

```
int main(){  
    fstream File;  
    File.open("A.txt",ios::in);  
  
    if(!File){  
        cout<<"Error";  
    }  
    else{  
        cout<<"Found the file";  
        File.close();  
    }  
    return 0;  
}
```

Code for
opening
a file

Explanation of the code

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We have a fstream library, which is responsible for handling files.



Creating an object of the fstream class and named it as 'File'.



On the above-created object, we have to apply the open() function to create a new file, and the mode is set to 'out' which will allow us to write into the file.



We use the 'if' statement to check for the file creation.



Prints the message to console if the file doesn't exist.



Prints the message to console if the file exists/created.



We use the close() function on the object to close the file.

Example

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```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    fstream file;
    file.open("C:\\Users\\Asad Computers\\Videos\\simple.txt",ios::in);

    if (!file.is_open()) {
        cout << "Error: File could not be opened!" << endl;
        //return 0; // file failed to open
    }
    else{
        cout <<"Found the file";
    }

    file.close(); // file opened successfully and now closed
    return 0;
}
```


Example using ifstream directly

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```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream file("C:\\Users\\Asad Computers\\Videos\\simple.txt");

    if (!file.is_open()) {
        cout << "Error: File could not be opened!" << endl;
        return 0; // file failed to open
    }

    file.close(); // file opened successfully and now closed
    return 0;
}
```

Code for creating a file

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```
fstream File;  
File.open("B.txt",ios::out);
```

Writing to a file

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- ▶ We will use stream insertion operator (<<) along with the text enclosed within the double-quotes.
- ▶ With open() function, we will create a new file named 'FileName' and then we will set the mode to 'ios::out' as we have to write the data to file

Code for writing the data

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```
#include<iostream>
#include<fstream>
using namespace std;
int main(){

    ofstream MyFile("FileName.txt");
    MyFile<<"Hello Filing"<<endl<<"we are doing good";

    MyFile.close();

    return 0;
}
```

Example

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```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream file;
    file.open("C:\\Users\\Asad Computers\\Videos\\output.txt", ios::out);

    if (!file.is_open()) {
        cout << "Error: Could not open file for writing!" << endl;
        return 0;
    }

    file << "This text is written using ios::out mode.\n";
    file << "It will overwrite the file if it already exists.";

    cout << "File written successfully!" << endl;
    file.close();
}
```

Tasks

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- ▶ Write code for writing to the file named `MyFile.txt`, with the help of `ofstream`.
- ▶ Write following message: "There are no secrets to success"

Solution

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```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream file("MyFile.txt");

    if (!file.is_open()) {
        cout << "Error: Could not open file!" << endl;
        return 0;
    }

    file << "There are no secrets to success";

    cout << "Message written successfully!" << endl;
    file.close();
}
```

Reading data from a file

- ▶ For reading the data from a file, we will have to assure that file must exist
- ▶ We will open the file in ios:in mode



```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    fstream file;
    file.open("C:\\Users\\Asad
Computers\\Videos\\simple.txt",ios::in);
    if(!file.is_open()){
        cout <<"File does not exist or recheck the
spelling" << endl;
        return 0;
    }
    string text;
    while (getline(file, text)) {
        cout << text << endl;
    }
}
```

Code for Reading the data

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    ifstream file("C:\\Users\\Asad Computers\\Videos\\sample.txt");

    if (!file.is_open()) {
        cout << "Error: Could not open file!" << endl;
        return 0;
    }

    string line;
    int lineNumber = 3; // line to read
    int count = 0;

    while (getline(file, line)) {
        count++;
        if (count == lineNumber) {
            cout << "Line " << lineNumber << ": " << line << endl;
            break;
        }
    }

    if (count < lineNumber) {
        cout << "File has less than " << lineNumber << " lines." << endl;
    }

    file.close();
}
```

Reading a particular line

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream file("C:\\Users\\Asad
Computers\\Videos\\sample.txt", ios::app);

    if (!file.is_open()) {
        cout << "Error: Could not open file for appending!" <<
endl;
        return 0;
    }

    file << "\\nThis line is appended to the file.";

    cout << "Line appended successfully!" << endl;
    file.close();
}
```

Code for Appending the Data

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int n = sizeof(arr)/sizeof(arr[0]);

    ofstream file("C:\\Users\\Asad Computers\\Videos\\array.txt",
ios::out);

    if (!file.is_open()) {
        cout << "Error: Could not open file!" << endl;
        return 0;
    }

    for (int i = 0; i < n; i++) {
        file << arr[i] << " "; // write elements separated by space
    }

    cout << "Array written to file successfully!" << endl;
    file.close();
}
```

Writing an array to the file

Tasks

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Create a function which takes an array of employee names as its parameter

The function stores each element of the array on a new line of the file named Emp_names.txt